

Some more preventive measures

1. Keep all applications, including operating system, updated with the latest patch holes.
2. Don't run applications with known security flaws.
3. Don't use file sharing programs.
4. Delete browser history and cookies after finishing browsing.
5. Turn-off computers and always-'on' Internet connections when not in use.
6. Use special caution when opening an e-mail attachment.
7. Be careful when downloading software for free.
8. Be careful when visiting a website and beware of parasite websites with almost the same name as the real website's.
9. Last but not the least, take regular backups of data on different media that are either kept unconnected to the system (most of the time), read-only or not accessible for other reasons, such as using different file systems. More importantly, keep the backups at a secure offsite location.

TCP splicing can prevent SYN floods. (SYN floods are used to launch denial-of-service attacks.) Automatic rate-limiting works as long as rate-thresholds are set accurately. Deep packet inspection can prevent bad content based attacks. Bogus IP filtering is helpful in defense against attacks originating from dark addresses or terminating to dark addresses.

Defense at router level

Router is a device that forwards data packets to the next network element on its journey by using information in its routing table or routing policy. Routers are very often attacked by intruders to gain access.

In order to safeguard the network, routers are also equipped with rate-limiting and access control list capabilities. These can be set either manually or by software control using network management systems. The purpose of access control is to authorise legitimate access, and to prohibit illegitimate access.

The essential notion is that without a legitimate access to an object, the system prohibits anything from happening with or to it. For this purpose, routers are equipped with access control lists that are the columns of the access matrix.

Access control list is a list of the users who are permitted access to the object, and those who are prohibited access to the object. It also lists the operations that are allowed on given objects. In this way, the control of access to the object is associated with

the object, which is an intuitively satisfying concept. Access control lists can generally be configured to control both inbound and outbound traffic. In this context, they are similar to firewalls.

Installation of application-front-end hardware

There are some intelligent hardware available in the market that can be placed on a network before traffic reaches the servers. Such hardware can be placed in addition to the routers and switches for added security. These hardware categorise data packets as regular, priority and dangerous, which helps in preventing packets having bad intent.

Sinkholing and blackholing

In blackholing technique, all the traffic of the attacked system is terminated to a null interface or non-existent server. Such technique has an advantage over classical firewalls since it is available on every potential network router, and adds virtually no performance impact.

Due to the nature of high-bandwidth routers, blackholing can often sustain higher throughput than conventional firewalls. For this reason, null routes are often used on high-performance core routers to mitigate large-scale denial-of-service attacks.

In a similar fashion, sinkholing is used to direct the traffic to a valid IP address where the traffic is analysed and segregated.

Clean pipes

The concept of using clean pipes includes passing all the traffic through a cleaning centre or a scrubbing centre via various methods like proxies, tunnels and direct circuit. At the cleaning centre, the traffic is segregated into good and bad traffic and only the good traffic is allowed to pass beyond the server. Central Internet connectivity is required for such kind of intrusion protection.

Analysing log files

Realising what has been happening on a system/network and anticipating what the hacker is going to try to penetrate the system/network can be done by a close examination of the log files. When a hacker is exploiting a program, his IP address is recorded in the log files. Even firewalls and routers have log files that can show evidence of hacking. Analysing log files also helps to ensure that everything is happening correctly. If something is found unlawful, preventive and corrective measures can be taken immediately before everything goes wrong and out of control.

Stack related countermeasures

Some hackers take advantage of buffer-overflow vulnerability. So in order to stop such instances, restrictions are put on buffer overflow. Moreover, it has been found that most applications do not use the stack for any type of execution. These stacks may be used to launch attacks. So stacks are made non-executable in order to block buffer overflows. This kind of defense is very helpful in stopping the insertion and use of shell code in an application. Another way to defense is to randomise the memory layout within the stack so that the hacker is unable to tell the location of the shell code implemented by the hacker.

Cryptographic techniques

Cryptography is the process of encrypting a plaintext with a mathematical algorithm, and the corresponding